Solution:

From the display, combine the stem with each of its leaves. The values are in the order from the smallest to the largest on the plot. Therefore, keep them in order and list the data values as follows:

252, 255, 260, 262, 276, 276, 276, 283, 289, 298

To determine the mean, add all the data values and then divide the sum by the number of data values.

$(252 + 255 + 260 + 262 + 276 + 276 + 276 + 283 + 289 + 298) \div 10$

$= 2727 \div 10 = 272.7$

Mean = 272.7.

The data is already arranged in ascending order. Therefore, identify the number in the middle position of the data. In this case, two data values share the middle position. To find the median, find the mean of these two middle data values.

The two middle numbers are 276 and 27 6.

The median is $(276 + 276) \div 2 = 276$.

The mode is the data value that occurs more frequently. Looking at the stem and leaf plot, we can see the data value 276 appears thrice.

Therefore the mode is 276.

Recall that the range is the difference of the greatest and least values. On the stem and leaf plot, the greatest value is the last value and the smallest value is the first value.

The range is 298 – 252 = 46.

Points to Remember

- Data array enables one to extract supplementary information from the data.
- It is essential that statistical data must be presented in a condensed form through classification.
- The process of dividing the data into different groups or classes which are as homogeneous as possible within the groups or classes, but heterogeneous between themselves is said to be classification

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